

MANTHAN

NEET 2027

ZOOLOGY

Lecture: 2

Body fluids and Circulation

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ZOOLOGY • MASTER TEACHER

VEDANTU • NEET 2027



TODAY'S GOAL

Blood - Plasma ✓

RBC ✓

WBC ✓



Student Details Form

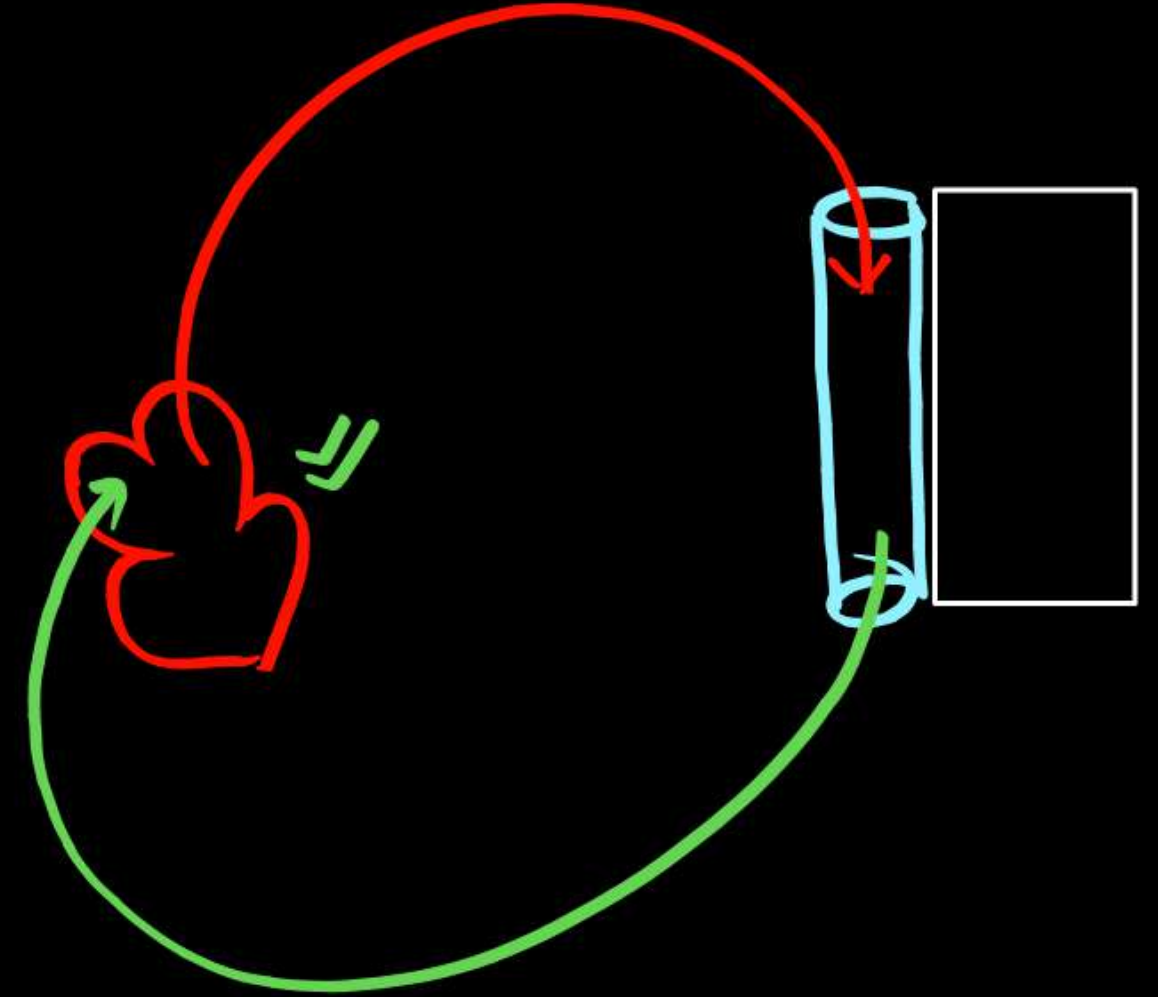


Telegram Channel

Dear students, Scan the QR codes to fill mandatory Student Details Form and join the official Telegram channel.

Circulatory system

1. Circulating fluid - Blood, Lymph
2. Blood vessels - Arteries, Capillaries, Veins and Portal System.
3. Pumping device – Heart





Blood

BLOOD

= Fluid Connective Tissue (origin - Mesodermal)

- Volume of blood -

Male - = 5 L ✓

Female - = 4.5 L ✓

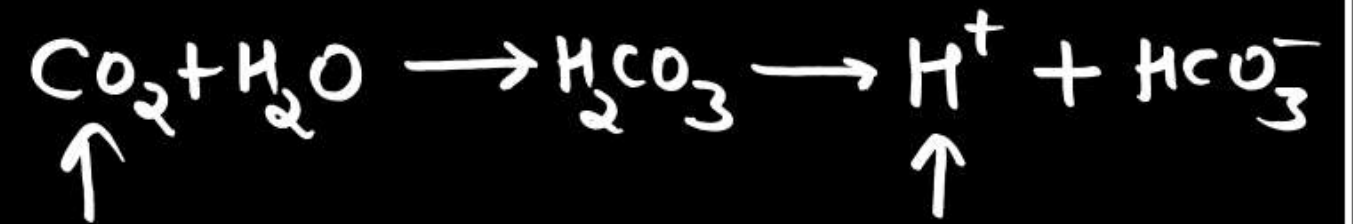
- ECF- = 15 L (Blood + lymph + CSF + Synovial fluid etc.)

- pH of blood -

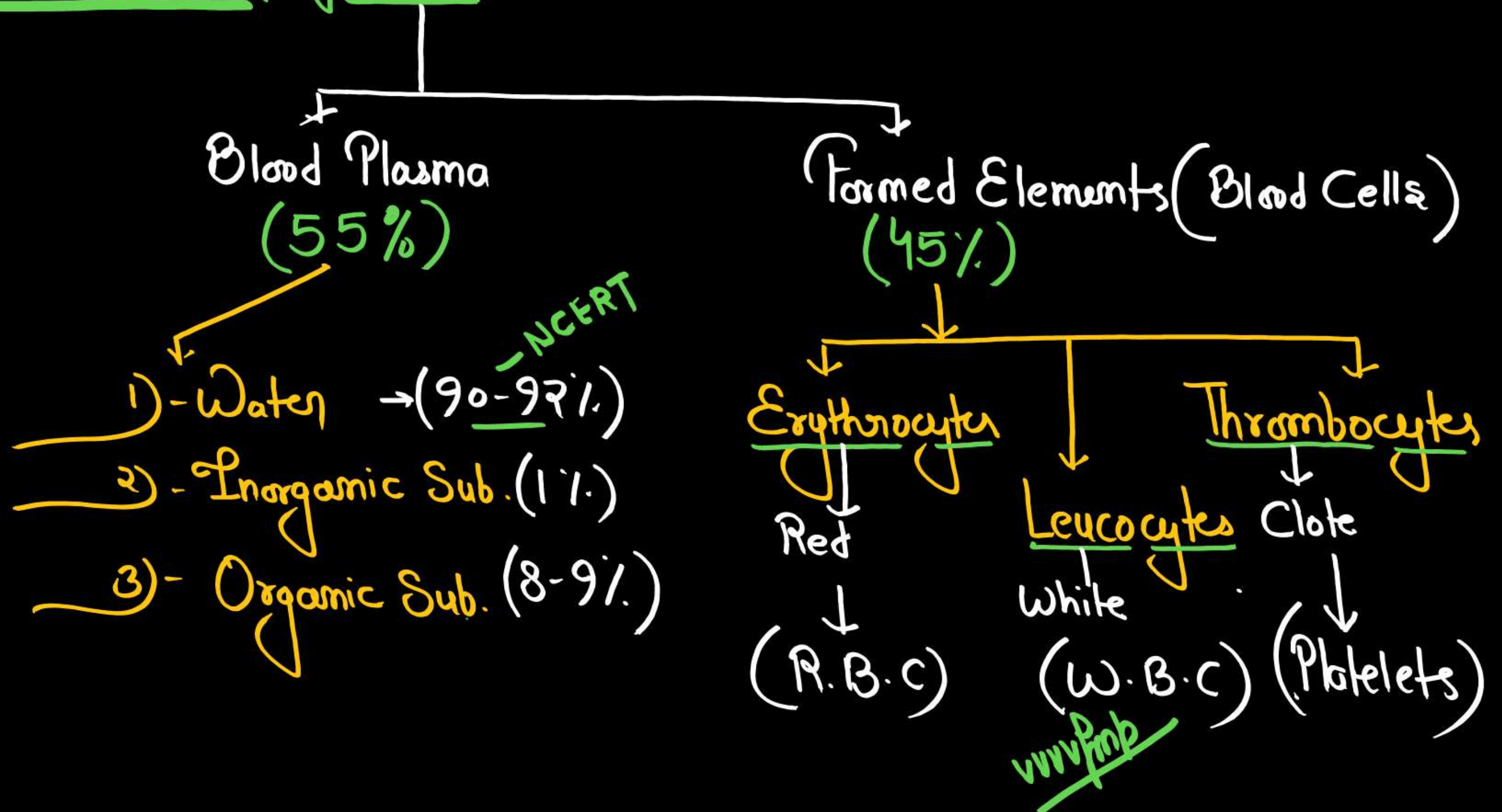
5 L (33% of ECF)

→ 7.4 [oxy. blood] = $\text{CO}_2 \downarrow \rightarrow \text{H}^+ \downarrow \rightarrow \text{pH} = \uparrow$ ✓

→ 7.2 [Deoxy. blood] = $\text{CO}_2 \uparrow \rightarrow \text{H}^+ \uparrow \rightarrow \text{pH} = \downarrow$ ✓

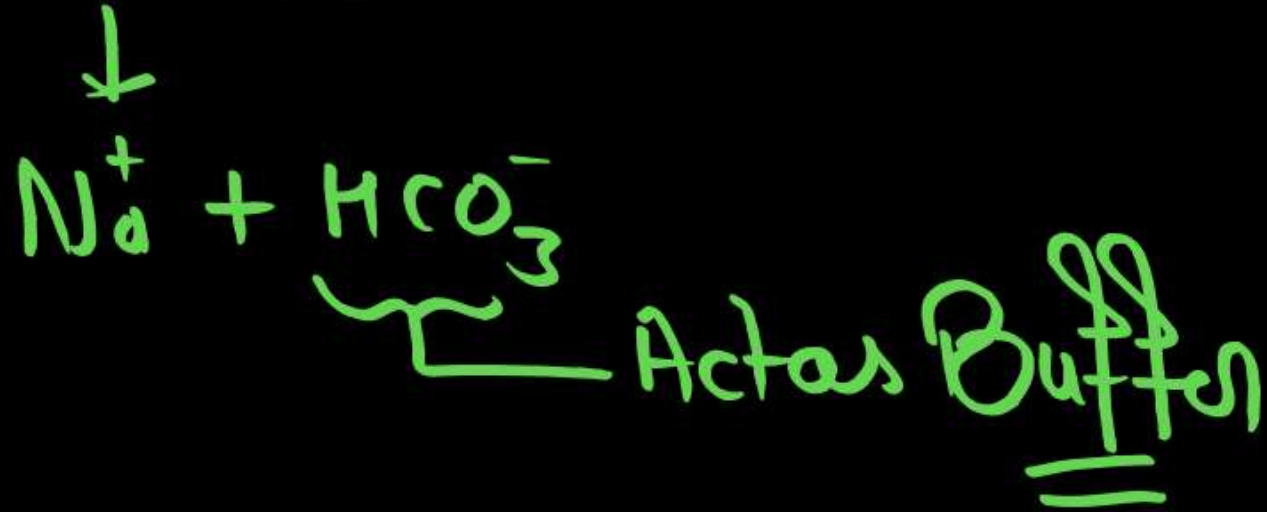
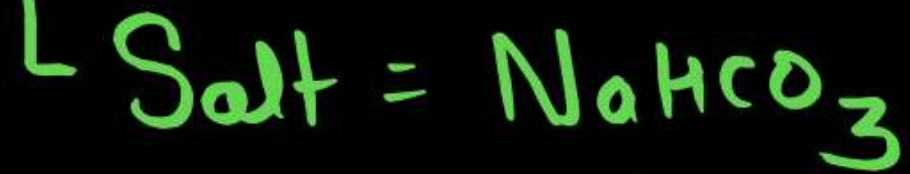
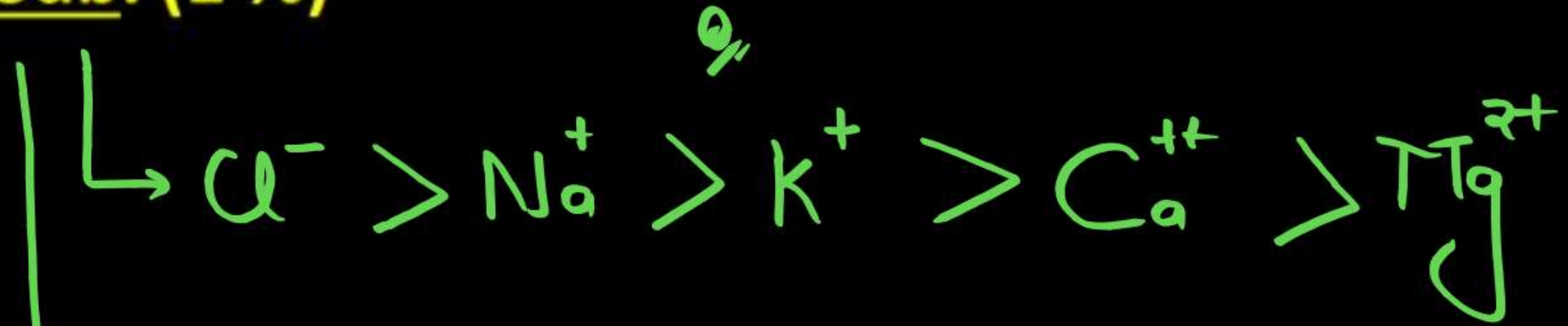


Composition of Blood →

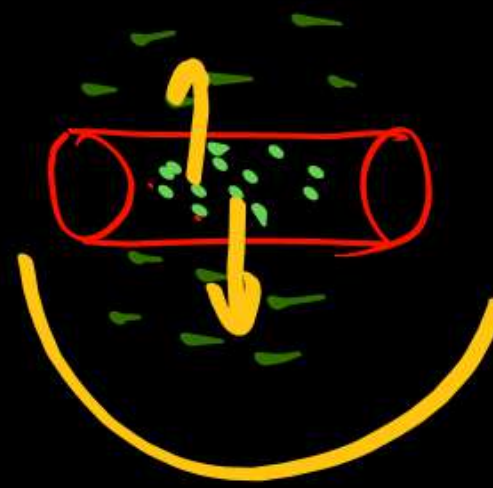


Plasma (55%)

(A) Inorganic Sub. (1 %)



(B) Organic Sub. (8-9 %) –



(1) Proteins (6-8 %) NCERT

Albumin

- Smallest blood protein
- Formed by Liver
- Function = Osmoregulation

Prothrombin
&
Fibrinogen

- Largest Blood Protein
- Formed by Liver
- Function = Blood Clotting

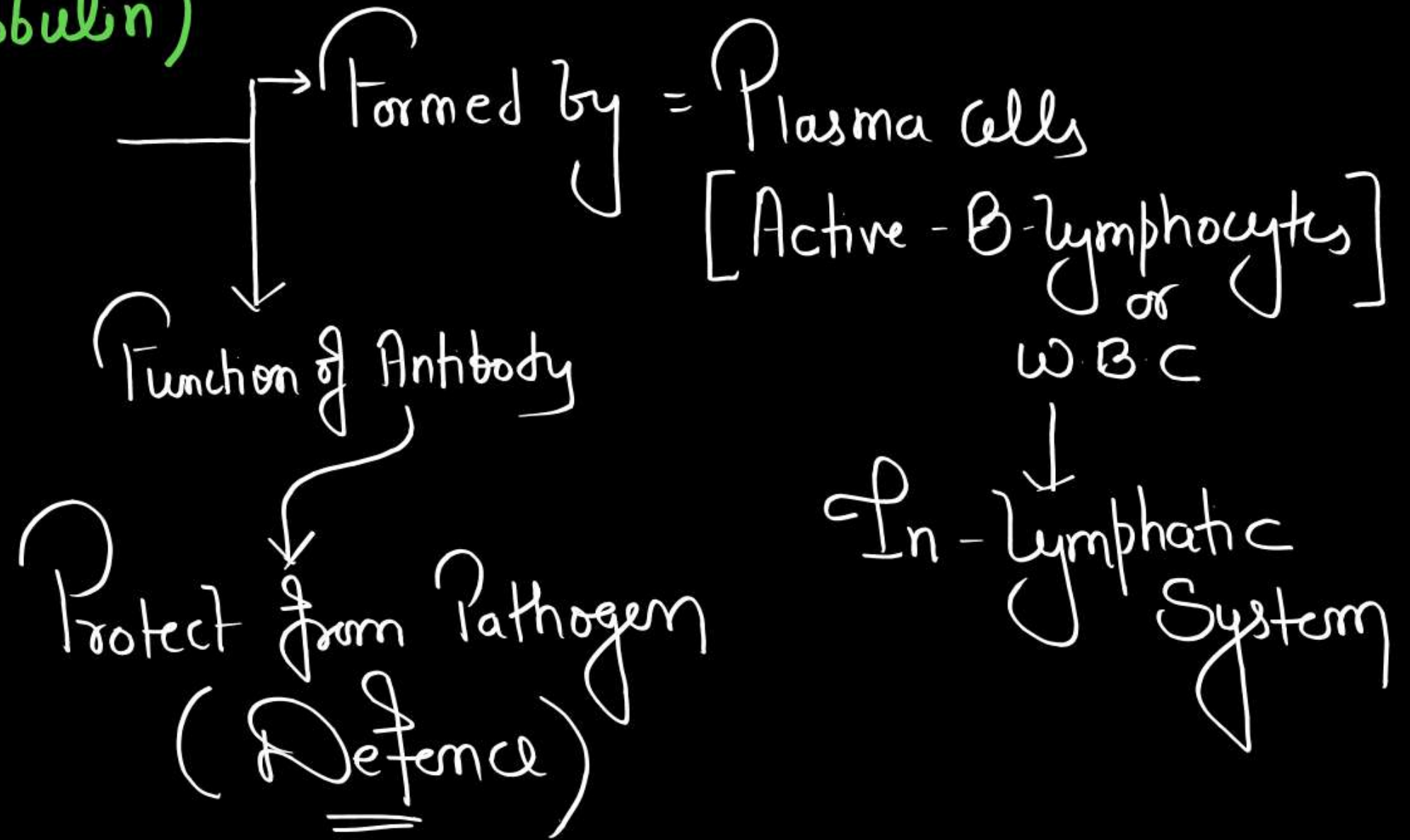
Globulin

α -globulin
 β -globulin
 γ -globulin

Formed by Liver
Binding & Transport of Vit. & Miners.



Immunoglobulin (γ globulin)
or
Antibody



2) - Blood Sugar $\rightarrow$ Normal = 100 mg/dL $\rightarrow$ if $\uparrow$ - Blood Glucose

Hyperglycemia $\rightarrow$ Diabetes + Ketosis

3) - Blood Cholesterol $\rightarrow$ Produced in Liver
Normal - 150 mg/dL
Value in Blood

4) - Blood urea $\rightarrow$ Normal Value = 20 - 40 mg/100 ml of blood
30 mg/100 ml

$\hookrightarrow$ if in urea in blood = Uremia

5) - Blood Hormones
etc.

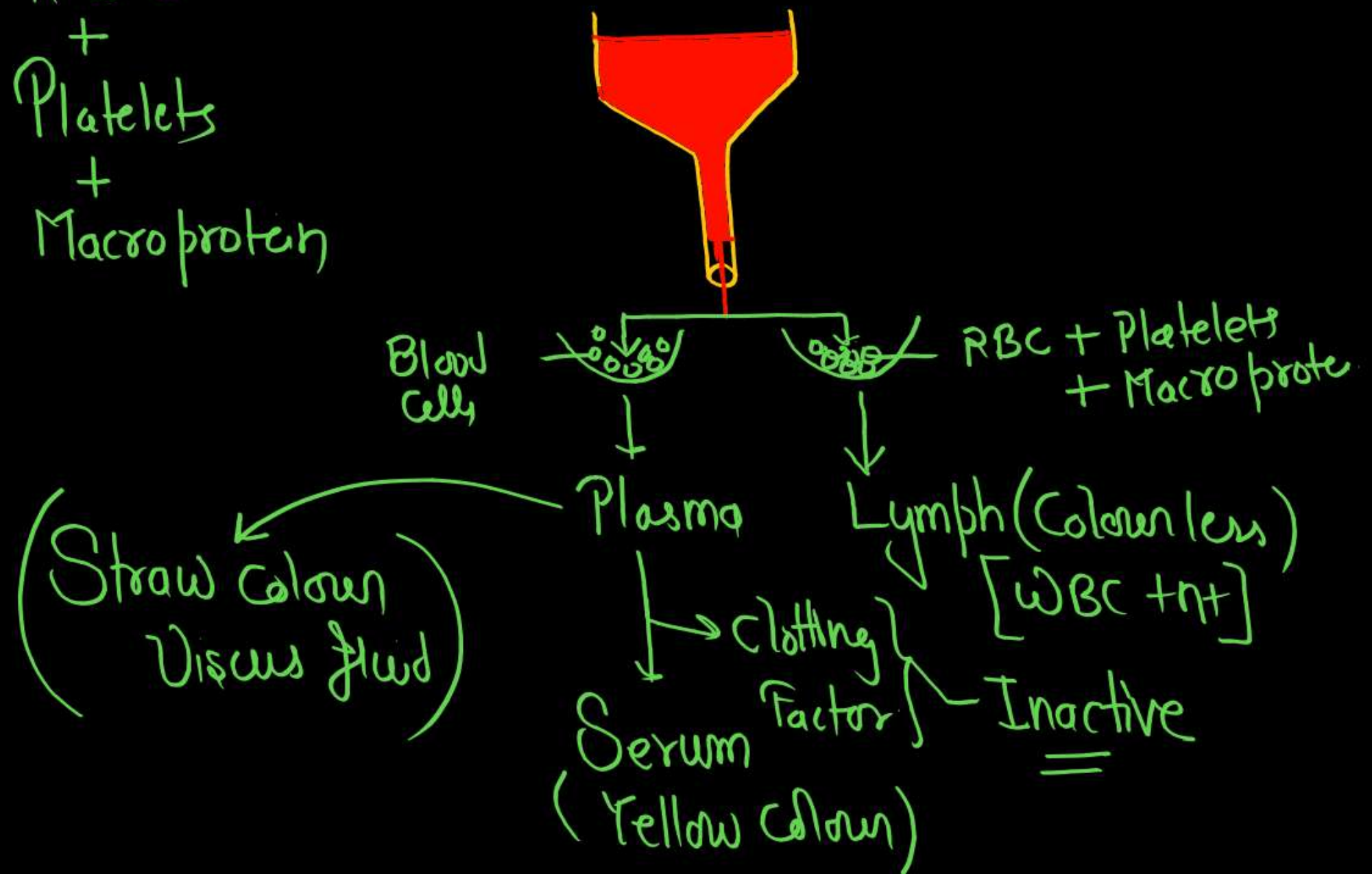
Note

Plasma = Blood — Formed Elements (Blood cells)

Serum = Plasma — Blood clotting factors

Lymph = Blood — RBC
+ Platelets
+ Macroprotein

or
Tissue
Fluid



Formed Elements (45 %)

(1) Erythrocytes (RBC)

Red

Cells

No. of RBC – In healthy human ≈ 5.5
million/mm³

Size and Shape of RBC -

In most of the

Mammals $\rightarrow$ RBC are

$\rightarrow$ Biconcave $\rightarrow$ $\uparrow$ Surface area

$\rightarrow$ Without Nucleus &
Cytoplasmic Organelles

} Reduced utilization
of O₂ in RBC

280 Millions
Molecules of
Haemoglobin

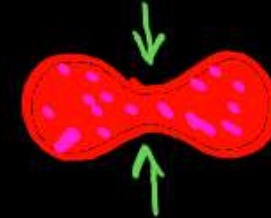
Metabolism + nt
or
Catabolism

$\rightarrow$ Glycolysis + nt

$\rightarrow$ Only - Anaerobic Respiration (Mitochondria - nt)

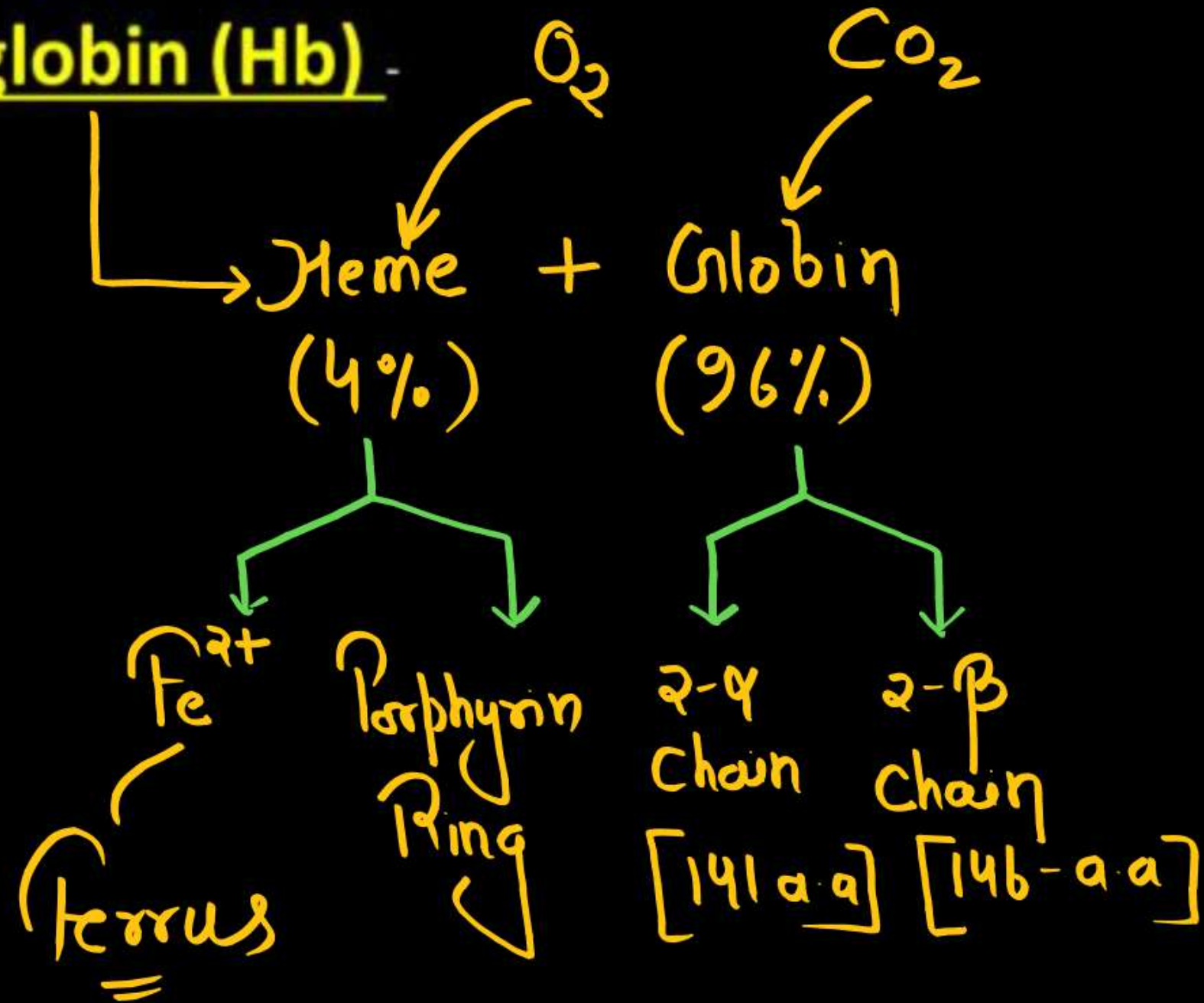


$\updownarrow$ 8 μ m



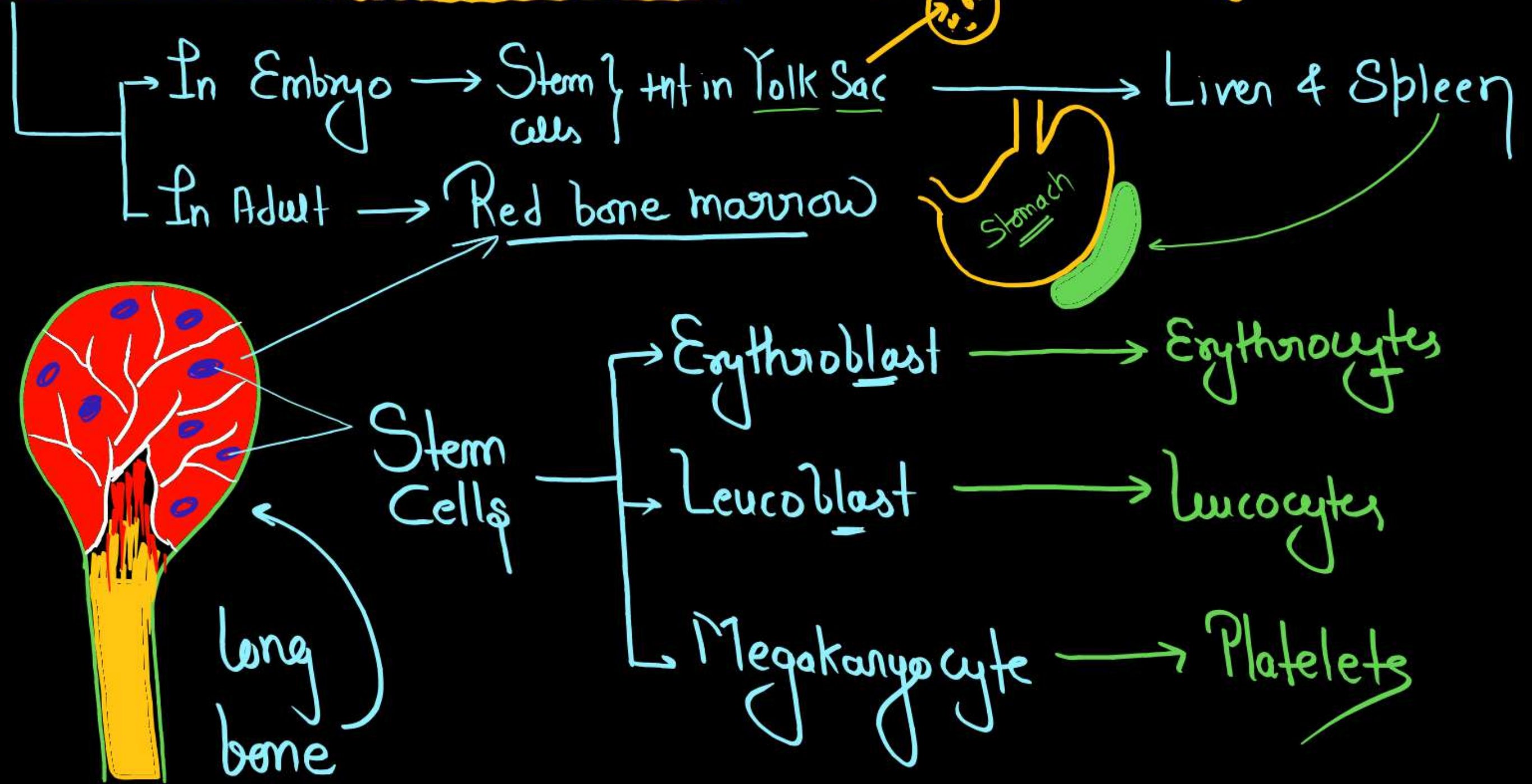
Biconcave

Haemoglobin (Hb)



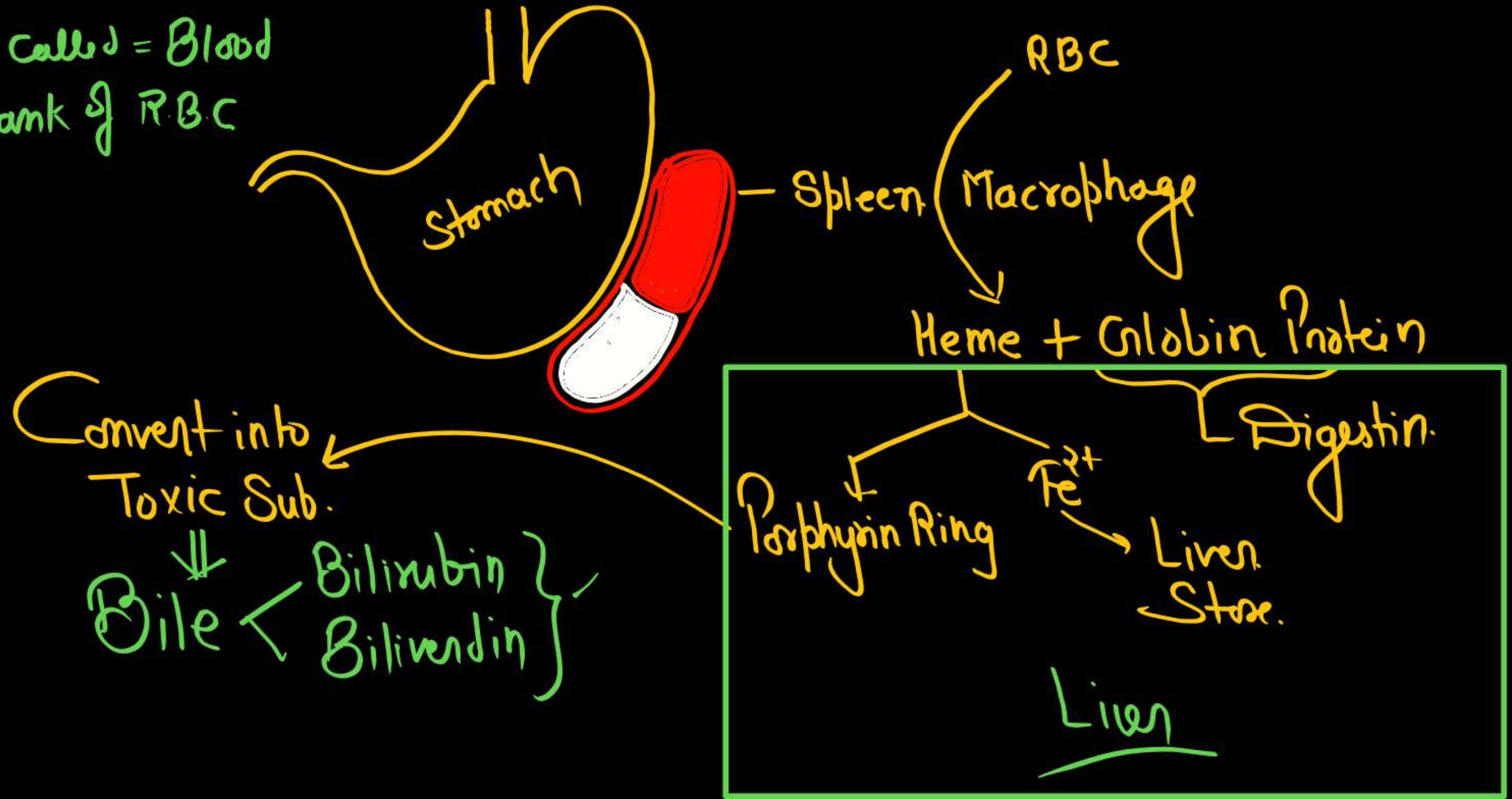
} 12-16 gm/100 ml of Blood

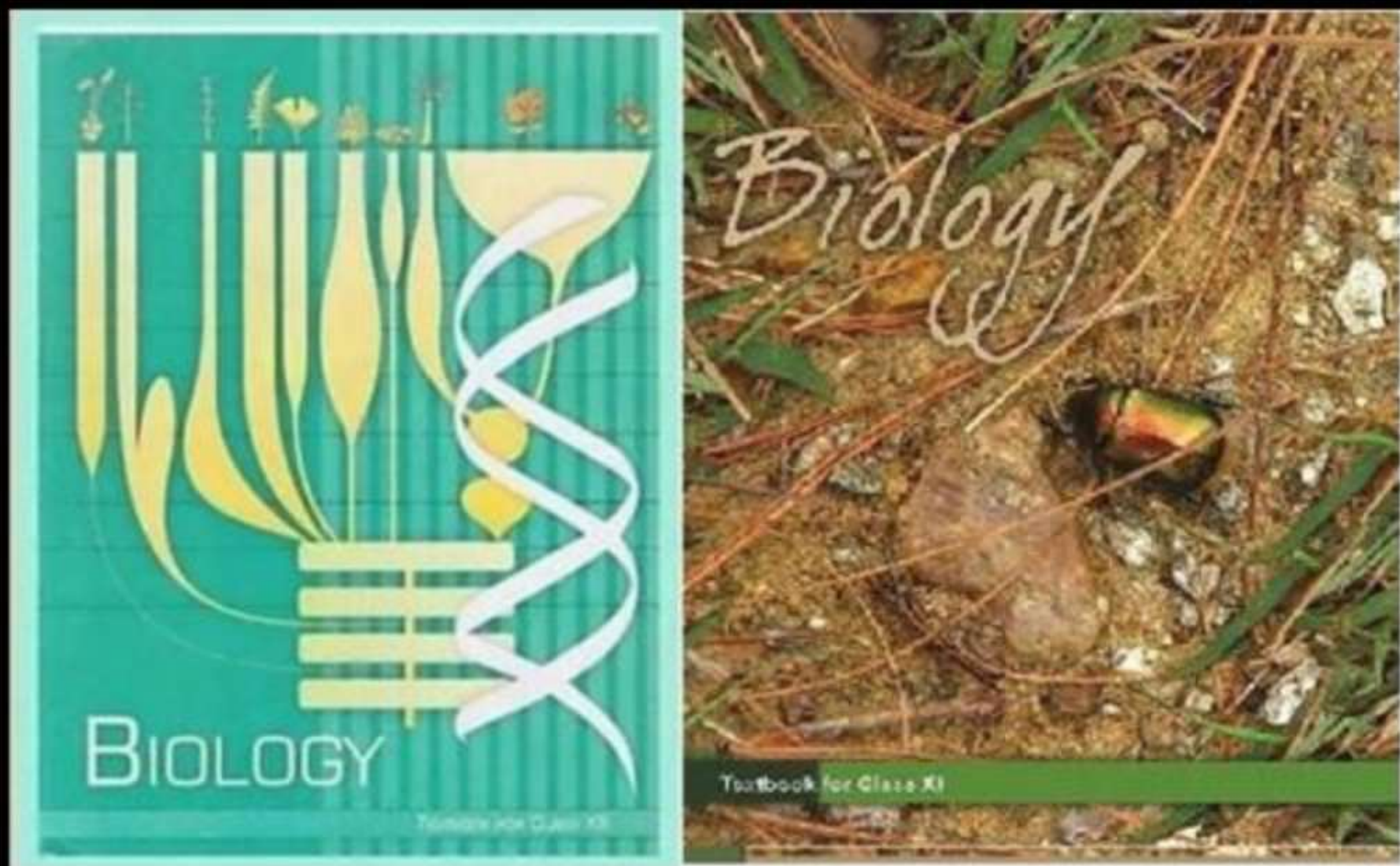
Formation of RBC (Erythropoiesis)



- Life Span of RBC = 120 days
- After 120 days – breakdown of RBC occur by macrophage in spleen etc. (Graveyard of RBC)

also called = Blood
Bank of RBC





NCERT line by line With Questions

BODY FLUIDS AND CIRCULATION

You have learnt that all living cells have to be provided with nutrients, O_2 and other essential substances. Also, the waste or harmful substances produced, have to be removed continuously for healthy functioning of tissues. It is therefore, essential to have efficient mechanisms for the movement of these substances to the cells and from the cells. Different groups of animals have evolved different methods for this transport. Simple

organisms like sponges and coelenterates circulate water from their surroundings through their body cavities to facilitate the cells to exchange these substances. More complex organisms use special fluids within their bodies to transport such materials. **Blood** is the most commonly used body fluid by most of the higher organisms including humans for this purpose. Another body fluid, **lymph**, also helps in the transport of certain substances. In this chapter, you will learn about the composition and properties of blood and lymph (tissue fluid) and the mechanism of circulation of blood is also explained herein.

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15.1 BLOOD

Blood is a special connective tissue consisting of a fluid matrix, plasma, and formed elements.

15.1.1 Plasma

Plasma is a straw coloured, viscous fluid constituting nearly 55 per cent of the blood. 90-92 per cent of plasma is water and proteins contribute 6-8 per cent of it. Fibrinogen, globulins and albumins are the major proteins.

Fibrinogens are needed for clotting or coagulation of blood. Globulins primarily are involved in defense mechanisms of the body and the albumins help in osmotic balance. Plasma also contains small amounts of minerals like Na^+ , Ca^{++} , Mg^{++} , HCO_3^- , Cl^- , etc. Glucose, amino acids, lipids, etc., are also present in the plasma as they are always in transit in the body. Factors for coagulation or clotting of blood are also present in the plasma in an inactive form. Plasma without the clotting factors is called serum.

15.1.2 Formed Elements

Erythrocytes, leucocytes and platelets are collectively called formed elements (Figure 15.1) and they constitute nearly 45 per cent of the blood.

Erythrocytes or red blood cells (RBC) are the most abundant of all the cells in blood. A healthy adult man has, on an average, 5 millions to 5.5 millions of RBCs mm⁻³ of blood. RBCs are formed in the red bone marrow in the adults.).

RBCs are devoid of nucleus in most of the mammals and are biconcave in shape. They have a red coloured, iron containing complex protein called haemoglobin, hence the colour and name of these cells. A healthy individual has 12-16 gms of haemoglobin in every 100 ml of blood. These molecules play a significant role in transport of respiratory gases. RBCs have an average life span of 120 days after which they are destroyed in the spleen (graveyard of RBCs)

HW

1. Blood, a special type of connective tissue

(A) Consists of a fluid matrix (Plasma)

(B) Has formed elements

(C) Is the most commonly used body fluid by most of the higher organisms

(D) All

2. Plasma is a straw coloured, viscous fluid constituting nearly ____ % of blood –

(A) 55

(b) 45

(C) 90

(D) 10

3. The amount of water present in blood plasma is –

(A) 99%

(B) 90-92%

(C) 10%

(D) 55%

4. Formed elements of blood include –

- (A) RBC, WBC and blood platelets
- (B) All solutes present in blood
- (C) Proteins present in blood
- (D) All minerals (elements)

5. Life span of human RBC is –

- (A) 120 hours
- (B) 120 month
- (C) 120 days
- (D) 102 days

6. What is the amount of haemoglobin present in 100 ml blood of human blood?

- (A) 45g
- (B) 18-20g
- (C) 12-16g
- (D) 10 -12g

(7) Mammalian RBCs are in shape-

- | | |
|---------------|-----------------|
| (A) Oval | (B) Biconvex |
| (C) biconcave | (D) Sickle like |

(8) Globulins contained in human blood plasma are primarily involved in

- (1) Osmotic balance of body fluids
- (2) Oxygen transport in the blood
- (3) Clotting of blood
- (4) Defense mechanisms of body

- (9) I. Plasma contains very high amount of minerals like Na^+ , Ca^{2+} , Mg^+ , HCO_3^- , Cl^-
II. Glucose, amino acids, lipids etc. are present in the plasma as they are not always in transit in the body
III. Factors for clotting of blood are present in plasma in an active form
IV. Plasma without the clotting factors is called serum

Find the correct one regarding the above statement

- | | |
|-------------------------|------------------------|
| (1) Only I is incorrect | (2) Only II is correct |
| (3) IV is correct | (4) All are incorrect |

Q. RBCs are formed in theA.... in the adults. RBCs are devoid of nucleus in most of the mammals and areB..... in shape. They have a red coloured, iron containing complex protein calledC....., hence the colour and name of these cells. A healthy individual hasD..... of haemoglobin in every 100 ml of blood.

- | | | | |
|--------------------------|---------------|-----------------|-----------------|
| (a) A = Red bone marrow, | B = Biconvex | C = Haemoglobin | D = 20 – 26 gms |
| (b) A = Liver, | B = Biconcave | C = Haemoglobin | D = 12 – 16 gms |
| (c) A = Red bone marrow, | B = Biconcave | C = Haemoglobin | D = 12 – 16 gms |
| (a) A = Liver, | B = Biconvex | C = Haemoglobin | D = 12 – 16 gms |

Q. Plasma is a straw coloured, viscous fluid constituting nearlyA... cent of the blood.

....B..... of plasma is water and proteins contribute.....C..... of it.

- | | | |
|----------------------|-----------------------|----------------------|
| (a) A = 55 per cent, | B = 6 – 8 per cent, | C = 90 – 92 per cent |
| (b) A = 55 per cent, | B = 90 – 92 per cent, | C = 6 - 8 per cent |
| (c) A = 20 per cent, | B = 6 – 8 per cent, | C = 90 – 92 per cent |
| (d) A = 55 per cent, | B = 6 – 8 per cent, | C = 10 – 12 per cent |

Q.A..... are needed for clotting or coagulation of blood.B.... primarily are involved in defense mechanisms of the body and theC..... help in osmotic balance

- | | | |
|---------------------|-----------------|----------------|
| (a) A = Albumin, | B = Fibrinogen, | C = Globulin |
| (b) A = Fibrinogen, | B = Albumin, | C = Globulin |
| (c) A = Globulin, | B = Albumin, | C = Fibrinogen |
| (d) A = Fibrinogen, | B = Globulin, | C = Albumin |

THANK YOU !!